MIL-S-19500/285A(EL)
6 February 1968

SUPERSEDING
MIL-S-19500/285(EL)
27 February 1964

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON
TYPES 1N660, 1N661

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for silicon, semiconductor diodes for use, particularly, as switching devices in relatively high-conductivity circuit applications. (See 6.2 herein.)

1.2 Outline and dimensions.- See Fig. 1 herein.

1.3 Operating characteristics and ratings.- (At T_A = +25°C, unless otherwise specified. See 3.4 herein.)

	V_R	I_O	i_f(surge) (at; 1/120 sec.)	Switching (t_{rr})	$T_{(oper.)}$		T_{stg}		Barom. Press.
	Max.	Max.	Max.	Max.	Min.	Max.	Min.	Max.	Max.
	Vdc	mAdc	mAdc	usec	°C		°C		(Alt.)ft.
1N660	-100	100	500	0.30	-65	+150	-65	+150	100,000
1N661	-200	100	500	0.30	-65	+150	-65	+150	100,000

FSC-5961

Source: https://assist.dla.mil -- Downloaded: 202610-18T14:04Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
MILITARY
MIL-S-19500 — Semiconductor Devices, General Specification For

STANDARDS
MILITARY
MIL-STD-15 — Electrical And Electronic Symbols
MIL-STD-750 — Test Methods For Semiconductor Devices

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements.- Requirements for the diodes shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and symbols.- The abbreviations and symbols used herein are defined in Specification MIL-S-19500 and as follows:

R_r . . . . . inverse recovery resistance

3.3 Design and construction.- The diodes shall be of the design, construction, and physical dimensions specified in Figure 1.

3.3.1 Terminal arrangement.- The terminal arrangement on the diodes shall be as indicated in Figure 1.

3.3.2 Operating position.- The diodes shall be capable of proper operation in any position.

Source: https://assist.dla.mil -- Downloaded: 202610-18T14:04Z
Check the source to verify that this is the current version before use.

3.4 Performance characteristics.- The diode performance characteristics shall be as specified in Tables I, II, and III herein. Except where specifically differentiated for a respective diode type (see 1.3 and Tables I, II, and III herein), the performance requirements including characteristics, ratings, and test conditions, apply equally to both diode types covered herein.

3.5 Marking.- Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the "classification" type designation only. The "manufacturer's identification" and "country of origin" may, at option of the manufacturer, be omitted from being marked directly on the semiconductor device covered herein. (See 3.5.1 herein.)

3.5.1 Polarity marking.- The graphic symbol for polarity indication on the semiconductor diode shall be as designated in Standard MIL-STD-15.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and acceptance inspection.- Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein (see 4.2.2 herein). Groups A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II and III, respectively, herein. Quality Conformance inspection shall include inspection of Preparation for Delivery (see 5.1 herein).

4.2.1 Specified LTPD for subgroups.- The LTPD specified for a subgroup in Tables I, II, and III herein shall apply for all of the tests, combined, in the subgroup.

Source: https://assist.dla.mil -- Downloaded: 202610-18T14:04Z
Check the source to verify that this is the current version before use.

4.2.2 Sampling and acceptance criteria for Qualification inspection.- For Qualification inspection, 30 sample units shall be subjected to all of the Group A, B and C tests in Table I, II, and III, respectively, under the following conditions of selection, test routine, and evaluation:

(a) The 30 sample units shall be selectable by the authorized Government representative from a minimum lot quantity of 100 units of the diode type concerned.

(b) Beginning with visual-mechanical examination (including dimensions), all of the selected 30 sample units shall be subjected in the sequential order indicated in Tables I, II, and III herein, to each test successively through all of the Group A, B, and C tests, except that the Salt Spray (Corrosion) test shall be the last test performed on the sample units. Throughout the sequence of tests, subgroup end-point tests shall be performed consistently as established in the Tables herein; however, the Lead Fatigue end-point test after Salt Spray (Corrosion) test shall not be applicable, and Life Test(s) duration may be 340 hours at option of the Government technical representative.

(c) For all 30 sample units tested, only one (1) failure will be permitted. Where a maximum of two (2) failures is found, an additional quantity of 30 sample units selected as specified in (a) above shall be subjected to the test routine specified in (b) above, and a maximum of one (1) failure in this additional sample quantity will be permitted.

4.2.3 Group C testing (Quality Conformance Inspection).- Group C tests shall be performed on a lot every 6 months. (See Table III herein.) The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of Group C, 6-month, test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required Group C tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

4.2.4 Disposition of sample units.- Sample units subjected to Group C, Subgroup 1 inspection shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroups 1, 2, 3, 4, 5 and 6 tests, and Group C, *subgroup 2 tests (these tests to be considered non-destructive)*, may be delivered on the contract or order provided that, after Group B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.3 Particular examination and test requirements.-

4.3.1 Interval for End-Point test measurements.- All applicable End-Point Test measurements shall be performed, after sample units have been subjected to required physical-mechanical or environmental test(s), in accordance with the following time-delay limitations:

(a) For Qualification inspection: within 24 hours.

(b) For Quality Conformance inspection: within 96 hours.

4.3.2 Mechanical damage resulting from tests.- Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit after any of the Group A, B, or C tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

Table I. Group A inspection.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		5				
2071	Visual and mechanical examination	---		---	---	---	---
	Subgroup 2		5				
4011	Forward voltage drop	I_F= 6 mAdc		V_F	---	1.0	Vdc
4016	Reverse current:	V_R= -10 Vdc					
	1N660			I_R	---	-1.0	uAdc
	1N661			I_R	---	-2.0	uAdc
4016	Reverse current:						
	1N660	V_R= -100 Vdc		I_R	---	-5	uAdc
	1N661	V_R= -200 Vdc		I_R	---	-10	uAdc
4021	Breakdown voltage:	I_R= 0.1 mAdc, max					
	1N660			BV	-110	---	Vdc
	1N661			BV	-220	---	Vdc
4001	Capacitance	v_r= -10 Vdc f = 1 mHz		C	---	2.7	uuf
4031	Reverse recovery time	Test Cond.A I_F= 30 mAdc V_R = -35 Vdc R_L = 2 K ohms ±10% C_L = 10 uuf R_r = 400 K ohms		t_{rr}	---	0.30	usec
2/	High-temperature operation:	T_C= +100° $^{+3°C}_{-0°}$					
	Reverse current:						
	1N660	V_R = -100 Vdc		I_R	---	50	uAdc
	1N661	V_R = -200 Vdc		I_R	---	100	uAdc

1/ See 3.4 herein.

2/ Test measurement(s) shall be made after thermal equilibrium has been reached at the temperature specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		10				
2066	Physical dimensions	---		---	---	---	---
	Subgroup 2		20				
2026	Solderability	---		---	---	---	---
1051	Temperature cycling	Test Cond.F 10 cycles					
1056	Thermal shock (glass strain)	Test Cond.A		---	---	---	---
1021	Moisture resistance	No initial conditioning		---	---	---	---
	End-point tests						
4011	Forward voltage drop	I_F = 6 mAdc		V_F	---	1.0	Vdc
4016	Reverse current:						
	1N660	V_R= -100 Vdc		I_R	---	-5	uAdc
	1N661	V_R= -200 Vdc		I_R	---	-10	uAdc
	Subgroup 3		20				
2016	Shock	Non-operating 500 G 5 blows of 1 msec ea. in orientations X1, Y1, Y2 (total = 15 blows)		---	---	---	---
2056	Vibration, variable frequency	10 G Orientations X1, Y1		---	---	---	---
2006	Constant acceleration (centrifugal)	20,000 G Orientations X1, Y1		---	---	---	---
	End-point tests: Same as for Subgroup 2 above						

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection -(Cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min.	Max.	Unit
	Subgroup 4		20				
2036	Terminal strength:						
	Tension	Test Cond.A. Weight = 2 lbs. t = 3 seconds		---	---	---	---
	Lead fatigue	Test Cond. E Weight = 1 lb. Max. Time ea. bend = 1 minute		---	---	---	---
	End-point tests:						
	Same as for Subgroup 2 above						
	Subgroup 5		20				
2/	Low-temperature operation:	$T_A = -65° \, ^{+0°}_{-5°}C$					
4016	Reverse current:						
	1N660	V_R= -100 Vdc		I_R	---	20	uAdc
	1N661	V_R= -200 Vdc		I_R	---	20	uAdc
4066	Surge current:	if(surge) = 500 mAdc 10 surges at 1 min. intervals		---	---	---	---
	1N660	v_r= 100 V					
	1N661	v_r= 200 V					
	End-point tests:						
	Same as for Subgroup 2 above						

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection -(Cont'd)

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 6		λ=10				
1026	Steady state operation life:	I_o=100 mAdc		---	---	---	---
	1N660	v = 70 Vac, 60 Hz					
	1N661	v = 140 Vac, 60 Hz					
	End-point tests:						
4011	Forward voltage drop	I_F = 6 mAdc		V_F	---	1.0	Vdc
4016	Reverse current:						
	1N660	V_R = -100 Vdc		I_R	---	-7	uAdc
	1N661	V_R = -200 Vdc		I_R	---	-15	uAdc

1/ See 3.4 and 4.3.1 herein.

2/ Test measurement shall be made after thermal equilibrium has been reached at the temperature specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

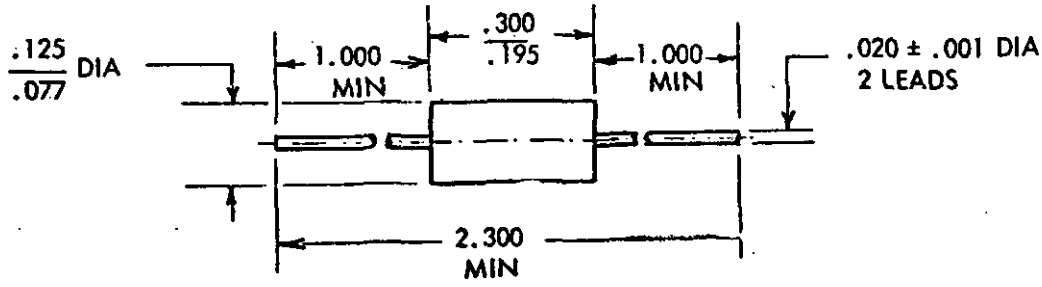
Table III. Group C inspection. 1/

Test Method per MIL-STD-750	Examination or test 2/	Conditions	LTPD	Symbol	Limits Min	Max.	Unit
	Subgroup 1		10				
1001	Barometric pressure, reduced (altitude operation):	Test Cond.D Energization duration= 1 minute					
	1N660	V_R= -100 Vdc		---	---	---	---
	1N661	V_R= -200 Vdc		---	---	---	---
1046	Salt spray (corrosion)	Test Cond. A					
	End-point tests:						
2036	Lead fatigue	Test Cond. E Weight = 1 lb. One arc		---	---	---	---
4011	Forward voltage drop	I_F = 6 mAdc		V_F	---	1.0	Vdc
4016	Reverse current:						
	1N660	V_R= -100 Vdc		I_R	---	-7	uAdc
	1N661	V_R= -200 Vdc		I_R	---	-15	uAdc
	Subgroup 2		λ=10				
1031	High-temperature life (non-operating)	T_{stg}= +150°C, min		---	---	---	---
	End-point tests:						
	Same as for Subgroup 1 above, but not including Lead Fatigue						

1/ See 4.2.3 herein.

2/ See 3.4 and 4.3.1 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.



All dimensions in inches.

Figure 1. Outline and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery, and the inspection of preparation for delivery, shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this document.

6.2 Application guidance.-

a. The revised requirements in this document issue do not affect replaceability between the diodes covered herein and the diodes, respectively, as covered in the previous issue(s) of this document.

b. To insure proper circuit application, particular attention should be given to the differential voltage-and-current requirements, ratings, and performance characteristics pertinent to the individual diode types covered herein.

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

6.4 Revision (document) changes.- Revision-letter symbols are not used in this document revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-EL

Preparing activity:
Army-EL

Project No. 5961-A115

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:04Z
Check the source to verify that this is the current version before use.